

$$\mathbb{E}[X] = \int_{-b}^b x f(x) dx$$

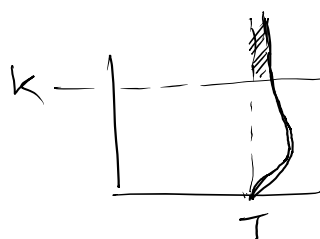
$$\frac{d}{dx} \left(\int_{a(x)}^{b(x)} f(x, t) dt \right) = \underbrace{f(x, b(x)) \left(\frac{db(x)}{dx} \right)}_{\text{upper limit}} - \underbrace{f(x, a(x)) \cdot \frac{da(x)}{dx}}_{\text{lower limit}} + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x, t) dt$$

$$C = e^{-rT} \cdot \mathbb{E}^Q \left[(S_T - k)^+ \right] = e^{-rT} \cdot \int_{\textcircled{k}}^{\textcircled{b_0}} \underbrace{(S_T - k)}_{\text{lower limit}} \cdot \underbrace{f(S_T)}_{\text{upper limit}} dS_T$$

$$\frac{\partial C}{\partial k} = \left[0 - \underbrace{(k - k)}_{\text{lower limit}} f(k) \times 1 + \int_{\textcircled{k}}^{\textcircled{b_0}} -f(S_T) dS_T \right] e^{-rT}$$

$$= -e^{-rT} \cdot \left\{ \int_{\textcircled{k}}^{\textcircled{b_0}} \underbrace{f(S_T)}_{\text{lower limit}} dS_T \right\}$$

$$= -e^{-rT} \times N(d_2)$$



$$\frac{\partial^2 C}{\partial k^2} = -e^{-rT} \times \left[0 - f(k) \cdot 1 \right] = e^{-rT} \cdot f_{S_T}(k)$$

Euler's homogeneous Function.

if $f(x, y)$ is a homogeneous function of degree 'n', then

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f \quad \checkmark$$

homogeneous function ??

$$f(\lambda x, \lambda y) = \lambda^{\boxed{n}} f(x, y) \quad \text{--- } \textcircled{n}$$

e.g. $f(x, y) = \underline{x^2 + y^2}$, $\underline{n=2}$

$$f(\lambda x, \lambda y) = \lambda^2 x^2 + \lambda^2 y^2 = \lambda^{\boxed{2}} f(x, y)$$

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 2f$$

$$x \cdot 2x + y \cdot 2y = 2f$$

$$c(s, k) = s N(d_1) - k e^{-rT} N(d_2)$$

$$\begin{aligned} c(\lambda s, \lambda k) &= \lambda s \cdot N(d_1) - \lambda k \cdot e^{-rT} \cdot N(d_2) \\ &= \lambda^{\boxed{1}} \cdot c \end{aligned}$$

$$\underline{s \frac{\partial c}{\partial s}} + \underline{k \frac{\partial c}{\partial k}} = c$$

$$c = s \frac{\partial c}{\partial s} + k \frac{\partial c}{\partial k}$$

$$\Rightarrow \frac{\partial c}{\partial s} = \frac{\partial c}{\partial s} + s \cdot \frac{\partial^2 c}{\partial s^2} + k \frac{\partial^2 c}{\partial k \partial s}$$

$$\Rightarrow s \frac{\partial^2 c}{\partial s^2} = -k \frac{\partial^2 c}{\partial s \partial k}$$

$$\Rightarrow s^2 \frac{\partial^2 c}{\partial s^2} = -k s \frac{\partial^2 c}{\partial s \partial k} = -k \frac{\partial}{\partial k} \left(s \frac{\partial c}{\partial s} \right)$$

$$= -k \frac{\partial}{\partial k} \left(c - k \frac{\partial c}{\partial k} \right)$$

$$= -k \frac{\partial c}{\partial k} - k \left[-k \cdot \frac{\partial^2 c}{\partial k^2} - \frac{\partial c}{\partial k} \right]$$

$$\left[s^2 \frac{\partial^2 c}{\partial s^2} = k^2 \frac{\partial^2 c}{\partial k^2} \right]$$